

CODE MODIFICATIONS & DOCUMENTATION

MAJOR CHANGES IMPLEMENTED

1. VIDEO INPUT SOURCE

Original: Live webcam feed (cv2.VideoCapture(0))

Updated: Video file input with configurable path

Benefits:

- Process pre-recorded videos for analysis
- Easier testing and debugging
- Reproducible results
- No hardware dependencies

2. TRACKING SYSTEM OVERHAUL

Original: Manual tracking using center-point distance calculation

Updated: YOLOv8 built-in tracker (model.track() with persist=True)

Improvements:

- More accurate ID assignment across frames
- Handles occlusions and crossing paths better
- No arbitrary distance thresholds
- Eliminates person_registry list management

3. AGE/GENDER DETECTION MODERNIZATION

Original: Caffe models (pre-2018 technology)

Updated: DeepFace library with modern deep learning

Comparison:

Old Approach:

- Required 4 external files (prototxt and caffemodel files)
- Limited accuracy on diverse demographics
- Age prediction in broad ranges: (0-2), (4-6), etc.
- Manual blob creation and preprocessing

New Approach:

- Single library installation (pip install deepface)
- State-of-the-art accuracy using VGG-Face, Facenet, etc.
- Precise age prediction (actual number)

- Automatic preprocessing and detection
- Gender output: Male/Female/Unknown

4. CONFIDENCE THRESHOLD FILTERING

Original: No confidence filtering (processes all detections)

Updated: Configurable CONFIDENCE_THRESHOLD parameter

Implementation:

- Set at 0.5 by default (50% confidence minimum)
- Passed directly to model.track(conf=CONFIDENCE_THRESHOLD)
- Reduces false positives significantly
- Easy to adjust based on use case

5. FRAME PROCESSING

Original: Frame skipping (frame_skip = 3, processes 1 out of 4 frames)

Updated: Processes every frame sequentially

Why This Matters:

- Better tracking continuity
- No missed detections
- Smoother output video
- More accurate CSV data

Trade-off: Slower processing, but more accurate results

6. OUTPUT VIDEO GENERATION

Original: Display only (no video saving)

Updated: Saves processed video with all annotations

7. RESOLUTION HANDLING

Original: Resizes frames to 416x416 before detection

Updated: Processes at original resolution

Impact:

- Better detection accuracy (especially for small objects)
- No coordinate scaling calculations needed
- More reliable bounding box positions

8. CODE ORGANIZATION

Original: ~180 lines with multiple toggle features

Updated: ~120 lines, streamlined and focused

Removed Features:

- Log file writing (redundant with CSV)
- Multiple detection mode toggles (o, p, f, a, b keys)
- Pause functionality
- Status panel with multiple indicators
- Frame resizing and ratio calculations

Configuration Section Added:

- All settings at the top of the file
- Clear variable names
- Easy to modify without searching through code